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Set 14: Survey Results (Unit IV)

Question 1: Stacked Bar Chart

Aim

To create a stacked bar chart showing the distribution of answers for Question 1 and label its elements.

Procedure

1. Load ggplot2 and create the survey data frame.
2. Count the answers of each question using table().
3. Draw a stacked bar of Question 1 answers with title and axis labels.

Code

```
library(ggplot2); library(gridExtra)
s <- data.frame(Respondent=1:3, Q1=c("A", "B", "C"), Q2=c("B", "A", "A"),
Q3=c("C", "D", "B"))
cnt <- as.data.frame(table(Question=rep(c("Q1", "Q2", "Q3"), each=3),
                             Answer=c(s$Q1, s$Q2, s$Q3)))
b <- ggplot(subset(cnt, Question=="Q1"), aes(Question, Freq, fill=Answer))
+
  geom_col() + labs(title="Distribution of Answers - Question 1",
x="Question", y="Count")
print(b)
```

Output

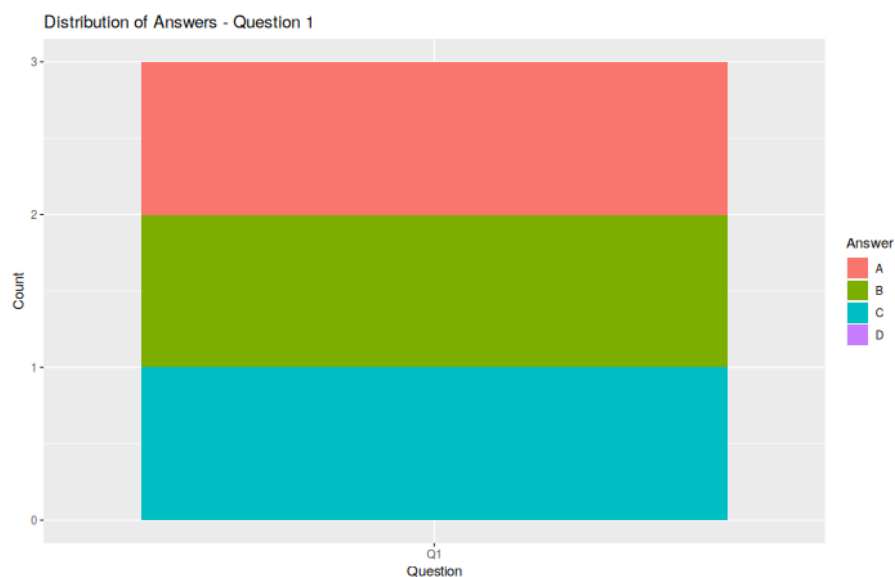


Fig 1: Stacked bar chart – Question 1

Result

The stacked bar chart was created. Answers A, B and C were each chosen once for Question 1 and D was not chosen.

Question 2: Radar Chart

Aim

To visualize the overall pattern of responses across all three questions using a radar chart.

Procedure

1. Use the answer counts (cnt) of Questions 1 to 3.
2. Draw each answer (A–D) as a polygon on a polar plot with straight edges.
3. Add title and legend.

Code

```
coord_radar <- function() { r <- coord_polar(); r$xis_linear <-  
function(coord) TRUE; r }  
r <- ggplot(cnt, aes(Question, Freq, group=Answer, color=Answer)) +  
  geom_polygon(fill=NA) + geom_point() + coord_radar() + ylim(0, 2) +  
  labs(title="Response Pattern (Q1-Q3)")  
print(r)
```

Output

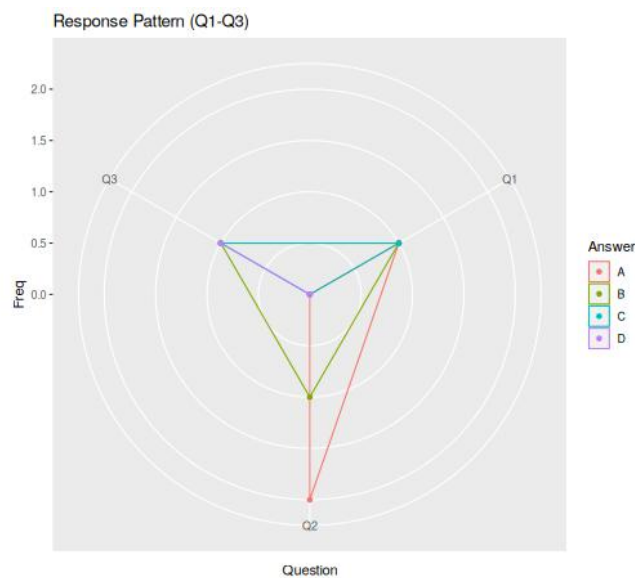


Fig 2: Radar chart – Questions 1 to 3

Result

The radar chart was created. Answer A is strongest on Question 2 (2 responses); B is chosen once on every question.

Question 3: Table

Aim

To build a labelled table showing the survey response data for each respondent.

Procedure

1. Print the survey data frame.
2. Draw it as a table using `tableGrob()`.

Code

```
print(s)
t <- tableGrob(s, rows=NULL)
grid.arrange(t)
```

Output

Console output:

```
Respondent Q1 Q2 Q3
1          1  A  B  C
2          2  B  A  D
3          3  C  A  B
```

Respondent	Q1	Q2	Q3
1	A	B	C
2	B	A	D
3	C	A	B

Fig 3: Survey response table

Result

The table of each respondent's answers to Questions 1 to 3 was created with labelled headings.

Question 4: Dashboard

Aim

To combine the stacked bar chart, radar chart and table into one dashboard for exploring the survey responses.

Procedure

1. Use the stacked bar (b), radar chart (r) and table (t) from the earlier questions.
2. Arrange all three in one panel using `grid.arrange()`.

Code

```
grid.arrange(b, r, t, ncol=2)
```

Output

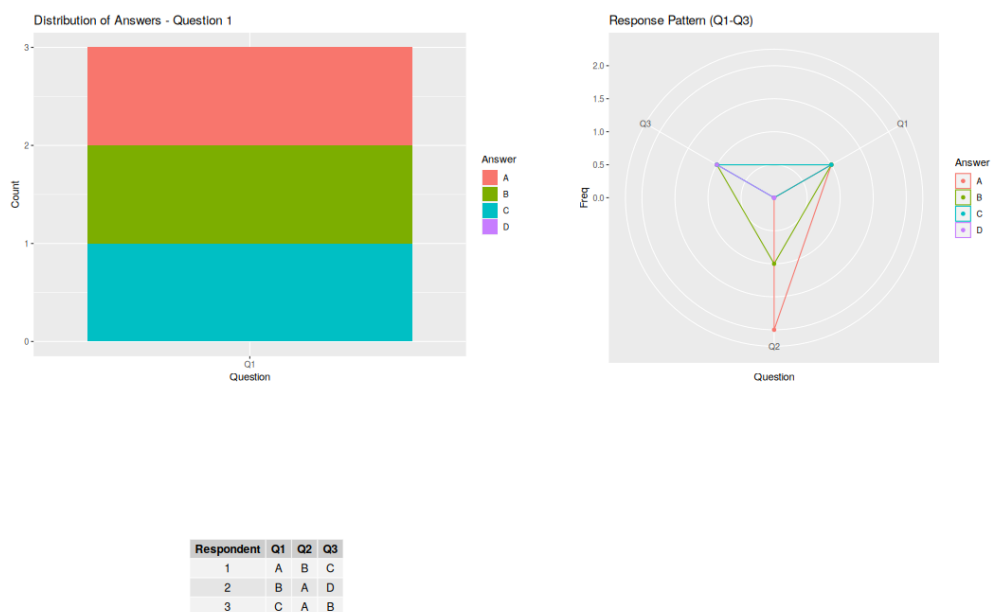


Fig 4: Dashboard (stacked bar + radar + table)

Result

The dashboard combining the stacked bar chart, radar chart and table was created in R.